

Limits – 8

Integer sequences: integrality, divisibility, periodicity

Homework

1 сентября 2026 года

Theoretical minimum. Linear recurrences with integer coefficients and their matrix form; conjugate pairs $\alpha^n + \beta^n$; periodicity of residues and the Pisano period; gcd properties of divisibility sequences; integrality of non-linear recurrences (relations of Somos type) and the trick of multiplying by the previous relation; the Skolem–Mahler–Lech theorem on the zeros of a linear recurrence (statement).

1. Prove that the sequence $a_{n+1} = \frac{a_n^2+2}{a_{n-1}}$ with $a_0 = a_1 = 1$ consists of integers and find a linear recurrence for it.
2. Prove that the Somos-4 sequence given by $a_n a_{n-4} = a_{n-1} a_{n-3} + a_{n-2}^2$ with initial values equal to 1 is integral, and explain the role played by the relation at the previous step.
3. Find the period of the Fibonacci sequence modulo 10 and modulo 11, and prove that the period modulo a prime p divides $p-1$ when $p \equiv \pm 1 \pmod{5}$ and divides $2(p+1)$ when $p \equiv \pm 2 \pmod{5}$.
4. Prove that $\lfloor (1 + \sqrt{2})^n \rfloor$ is odd for every positive integer n , and find a formula for $(1 + \sqrt{2})^n + (1 - \sqrt{2})^n$.
5. Let a_n be the integer sequence with $a_{n+1} = 3a_n - a_{n-1}$, $a_0 = 0$, $a_1 = 1$. Prove that a_m divides a_n if and only if m divides n , and that consecutive terms are coprime.
6. State the Skolem–Mahler–Lech theorem and test it on $a_n = 2^n - 3^n + 1$: find all n with $a_n = 0$.

7. Prove that if an integral sequence satisfies $a_{n+1} = \frac{a_n^2 + c}{a_{n-1}}$ with integer initial values, then for a suitable c all terms are integers; describe all such c for $a_0 = a_1 = 1$.
8. (ICMC 2023–24) Let $t_1 = 1$, $t_{2k} = -t_k$ and $t_{2k+1} = t_{k+1}$ for $k \geq 1$. Prove that the series $\sum_{n \geq 1} \frac{t_n}{n^{1/2024}}$ converges to a positive real number.
9. Prove that $\lfloor (2 + \sqrt{5})^n \rfloor$ is divisible by 2^n for odd n .
10. Let (a_n) be an integral sequence with $a_n \mid a_{n+1}$ and $\frac{a_{n+1}}{a_n}$ bounded. Prove that the ratio is eventually constant, and give an example showing that boundedness cannot be dropped.